Calculation Supplement — Document 4

Fisher, Chi-Square, and Cramér’s V

Nine-method mission-success analysis

343 runs / 9 methods / 4 scenarios

Step-by-step statistical analysis
of social navigation in an agricultural digital twin

Purpose. To support the calculations of success rates, scenario-wise Fisher exact tests, the global 9×2 chi-square test, and Cramér.s V.

Contents

1 Foundation

For a 2×2 table, Fisher uses

$$P(a) = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{a+b+c+d}{a+c}}. \tag{1}$$

Success rate is $\eta_s = N_{success}/N_{planned}$. Fisher conditions on fixed margins and, for a two-sided test, sums the observed table and all tables whose probability is no greater than the observed probability.

2 Calculation procedure

1. Classify each run as success or failure using completion of $A \to B \to A$ within 180 s.
2. Build a scenario-specific 2×2 success/failure table for each method pair.
3. Compute the exact hypergeometric probability and sum equally or more extreme tables.
4. Build the global method-by-outcome table, compute E_{ij}, and sum Pearson contributions.
5. Compute V to quantify association strength independently of significance.

For M4 versus M1 in E4,

	Success	Failure
$M4$	10	0
$M1$	2	8

and the exact two-sided sum gives $p = 7.14 \times 10^{-4}$. For the global table, $E_{ij} = R_i C_j / N$ and

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}. \tag{2}$$

3 Success rates

Table 1: Strict $A \to B \to A$ mission within 180 s.

Method	E1	E2	E3	E4	Global
M0	100%	100%	100%	100%	100% (40/40)
M1	100%	100%	100%	20%	80% (32/40)
M2	100%	100%	100%	20%	80% (32/40)
M3	100%	100%	100%	100%	100% (40/40)
M4	100%	100%	100%	100%	100% (40/40)
B1	90%	100%	100%	78%	92.3% (36/39)
B2	100%	100%	100%	100%	100% (40/40)
B3	100%	100%	100%	100%	100% (40/40)
B4	100%	80%	100%	90%	92.5% (37/40)

4 Statistical tests

In E4, M4 versus M1 yields $(10, 0; 2, 8)$ and $p = 7.14 \times 10^{-4}$; M4 versus M2 is identical. The global nine-method contingency gives $\chi^2 = 40.1$, $df = 8$, $p = 3.1 \times 10^{-6}$, and $V = 0.334$. Restricted to E4: $\chi^2 = 55.3$, $V = 0.788$. M4 versus B1 and B4 in E4 is nonsignificant: $p = 0.21$ and $p = 1.0$.

5 Conclusion

M1/M2 exhibit severe E4 deadlock; continuous-action methods retain $\eta_s \geq 92\%$. B1/B4 shortfalls are mild liveness losses, not binary-supervisor hard deadlock..

6 Reproducibility control

Method-specific denominators must be preserved: B1 has 39 valid runs; the remaining global denominators are shown in the table. Fisher is not replaced by an asymptotic approximation for sparse cells. Rates are calculated from integer counts before percentage rounding.